Equations of lines

1

a Yes

b No

c No

d Yes

2

a

x	-1	0	1	2
y	-1	1	3	5

b $m = 2, c = 1$

c $y = 2x + 1$

d $(21, 43)$

3

a

x	-1	0	1	2
y	7	4	1	-2

b $m = -3, c = 4$

c $y = -3x + 4$

d $(27, -77)$

4

a

x	-2	-1	0	1
y	-7	-5	-3	-1

b $m = 2, c = -3$

c $y = 2x - 3$

d $y = 77$

5

a

x	-1	0	1	2
y	0	-1	-2	-3

b $m = -1, c = -1$

c $y = -x - 1$

d $y = -20$

6

a

x	-5	0	5	10
y	-3	-2	-1	0

b 1

c $\frac{1}{5}$

x

d $m = \frac{1}{5}, c = -2$

e $(25, 5)$

7

a

x	-5	0	5	10
y	3	2	1	0

b $m = -\frac{1}{5}, c = 2$

c $y = -\frac{x}{5} + 2$

d $(-45, 11)$

8

a

x	-5	0	5	10
y	-6	-3	0	3

b $m = \frac{3}{5}, c = -3$

c $y = \frac{3x}{5} - 3$

d $(45, 24)$

9

a

x	0	1
y	1	3

b 2

c $m = 2, c = 1$

d $y = 2x + 1$

e $y = 45$

10

a

x	0	1
y	2	-2

b 4

c $m = -4, c = 2$

d $y = -4x + 2$

e $y = -106$

11

a 2

b $m = 2, c = -3$

c $y = 2x - 3$

d $(50, 97)$

12

a 1

b $\frac{1}{2}$

c $m = -\frac{1}{2}, c = 1$

d $y = -\frac{x}{2} + 1$

e $(-14, 8)$

13

a

i $(0, 1)$

ii 3

iii $y = 3x + 1$

b

i $(0,4)$ ii -2 iii $y=-2x+4$

c

i $(0,-1)$ ii $\frac{1}{2}$ iii $y=\frac{x}{2}-1$

d

i $(0,2)$ ii $-\frac{4}{3}$ iii $y=-\frac{4x}{3}+2$

14

a $y=-2x+4$ b $y=\frac{x}{4}-1$ c $y=\frac{5x}{4}-3$ d $y=-\frac{3x}{2}+1$

15

a $y=2x+1$ b $(22,45)$

16

a $y=\frac{2x}{5}-1$ b $(25,9)$

17

a $y=-3x+4$ b $y=9x-2$ c $y=-x+7$ d $y=\frac{3}{4}x$

e $y=3x$ f $y=-5x$ g $y=6x-5$ h $y=2x+2$